

Chapter 4.

4.1. Try to find the frequency spectrum of the periodic signal shown in figure 4-1.

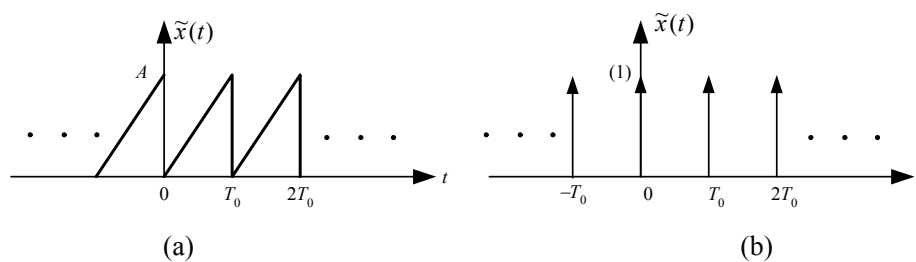


Figure.4-1

4.2. Known the frequency spectrum of continuous periodic signals shown in figure 4-2, try to find periodic signals $\tilde{x}(t)$ ($\omega_0 = 3$).

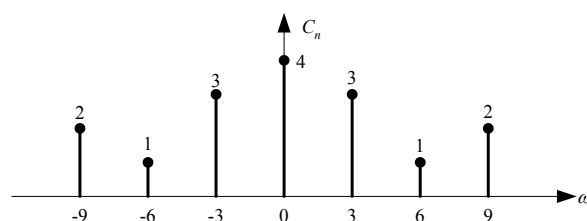


Figure.4-2

4.3. Figure 4-3 shows four periodic rectangular signals. Try to calculate the spectrum by symmetrical characteristic.

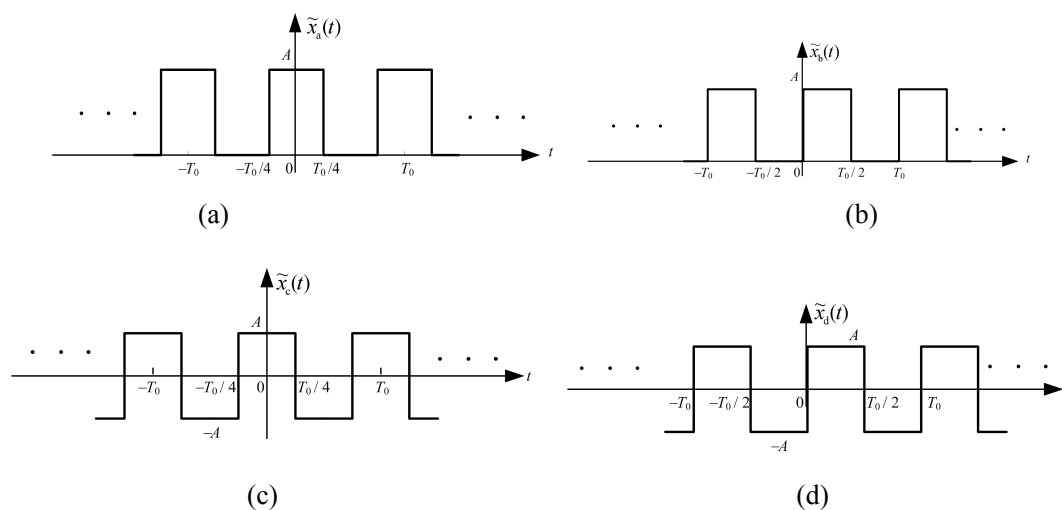


Figure.4-3

4.4. We know the periodic signal is $\tilde{x}(t) = 2\cos(2\pi t - 3) + \sin(6\pi t)$, try to sketch its frequency spectrum and power spectrum, and calculate its average power P .

4.5. Try to write the spectrum function $X(j\omega)$ of the following signals. Where ω_0 is a constant.

(1) $x(t) = \cos[\omega_0(t - t_0)]$

(2) $x(t) = \sin^2(\omega_0 t) \cdot u(t)$

(3) $x(t) = u(t) - u(t - 2)$

(4) $x(t) = e^{-2t} [u(t) - u(t - 2)]$

$$(5) x(t) = e^{-3|t-2|}$$

$$(6) x(t) = \frac{d}{dt} (te^{-2t} \sin(t) u(t))$$

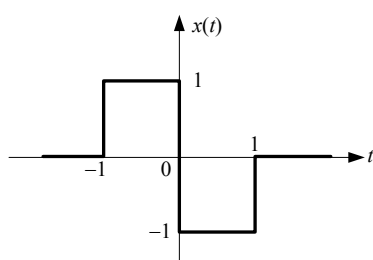
$$(7) x(t) = \int_{-\infty}^t \frac{\sin(\pi x)}{\pi x} dx$$

$$(8) x(t) = \left[\frac{2 \sin(\pi t)}{\pi t} \right] \left[\frac{2 \sin(2\pi t)}{\pi t} \right]$$

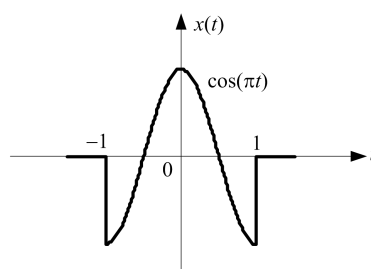
$$(9) x(t) = \frac{d}{dt} \left[\frac{\sin(\pi t)}{\pi t} * \frac{\sin(2\pi t)}{\pi t} \right]$$

$$(10) x(t) = \int_0^t e^{-3\tau} e^{-2(t-\tau)} d\tau, t \geq 0$$

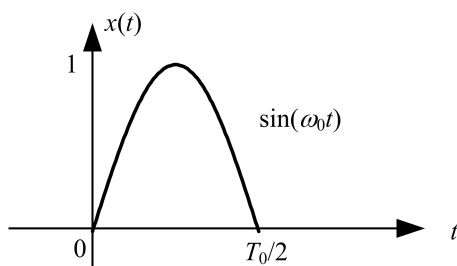
4.6. Try to find the frequency spectrum of signals shown in figure 4-4 by the properties of fourier transform.



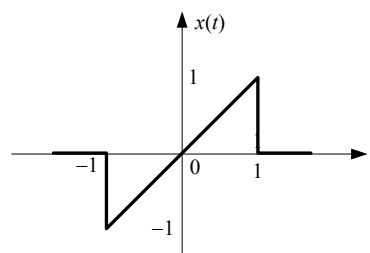
(a)



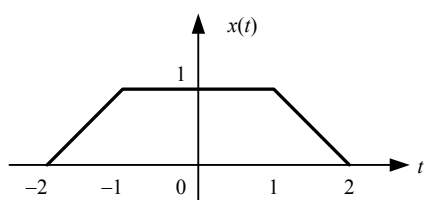
(b)



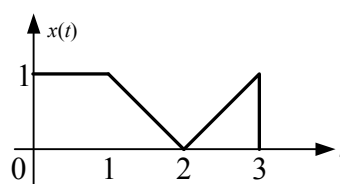
(c)



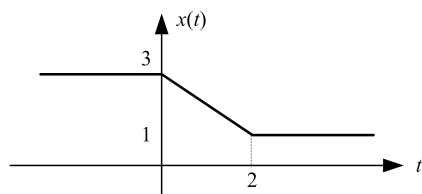
(d)



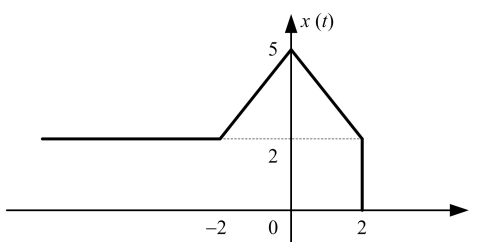
(e)



(f)



(g)



(h)

Figure.4-4

4.7. Try to find the frequency spectrum function of the following signals using the reciprocity

symmetry property of fourier transform.

$$(1) x(t) = \frac{\sin \pi t}{t} \quad (2) x(t) = \frac{1}{a^2 + t^2} \quad (3) x(t) = \frac{1}{a + jt} \quad (4) x(t) = \frac{1}{\pi t}$$

4.8. Try to find the signals $x(t)$ corresponding to the following spectrum functions.

$$\begin{aligned} (1) X(j\omega) &= \frac{3}{j\omega + 2} + \frac{4}{j\omega - 2} & (2) X(j\omega) &= \frac{1}{j\omega(j\omega + 1)} + 2\pi\delta(\omega) \\ (3) X(j\omega) &= \text{Sa}^2(\omega\tau) & (4) X(j\omega) &= -\frac{2}{\omega^2} \\ (5) X(j\omega) &= \frac{2\sin(\omega)}{\omega(j\omega + 1)} & (6) X(j\omega) &= \frac{4\sin(2\omega - 2)}{2\omega - 2} + \frac{4\sin(2\omega + 2)}{2\omega + 2} \\ (7) X(j\omega) &= \frac{d}{d\omega} \left[4\cos(3\omega) \frac{\sin(2\omega)}{\omega} \right] & (8) X(j\omega) &= \text{Im} \left[e^{-j3\omega} \frac{1}{j\omega + 2} \right] \end{aligned}$$

4.9. We know signal frequency spectrum $X(j\omega)$ shown in figure 4-5, find the signal $x(t)$.

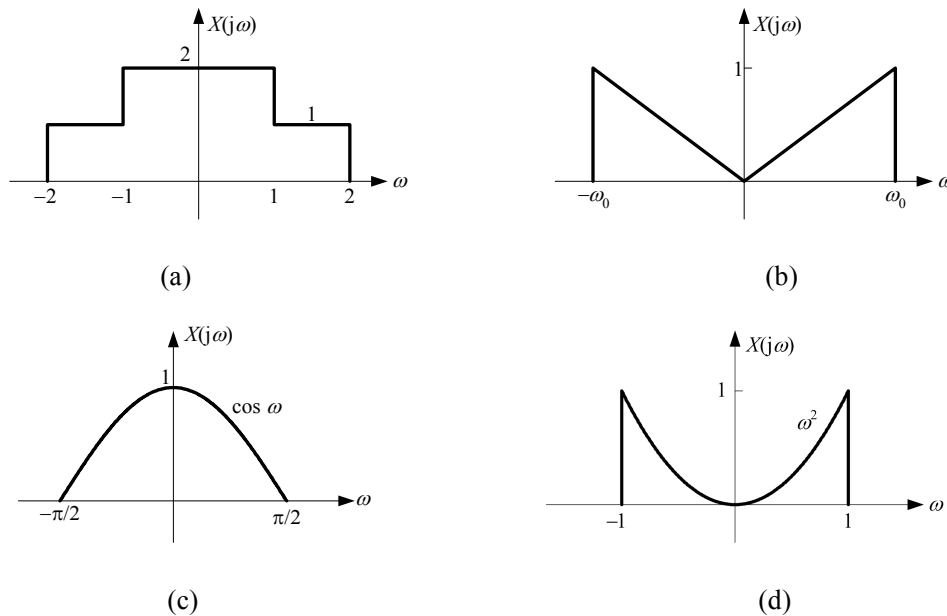


Figure 4-5

4.10. In the system shown in figure 4-6, the periodic signal $p(t)$ is a circular rectangular signal with a width τ ($\tau < T_0$). The frequency spectrum of signal $x(t)$ is $X(j\omega)$.

- (1) Try to calculate the frequency spectrum C_n of periodic signal $p(t)$.
- (2) Calculate frequency spectrum density function $P(j\omega)$ of signal $p(t)$.
- (3) Try to find frequency spectrum expression $X_p(j\omega)$ of signal $x_p(t)$.
- (4) If the signal $x(t)$ the highest frequency ω_m is known, in order to make $X_p(j\omega)$ spectrum aliasing, what is the maximum advisability of T_0 .

4.11. Try to determine the frequency spectrum $\tilde{X}[m]$ of the following sequence $\tilde{x}[k]$ with the cycle of 4.

$$(1) \tilde{x}[k] = \{\cdots, \overset{\downarrow}{1}, 2, 0, 2, \cdots\}, \quad (2) \tilde{x}[k] = \{\cdots, \overset{\downarrow}{0}, 1, 0, -1, \cdots\}$$

4.12 Try to determine the period N and frequency spectrum $\tilde{X}[m]$ of the following periodic sequences.

$$(1) \tilde{x}[k] = \sin(\pi k / 4) \quad (2) \tilde{x}[k] = 2\sin(\pi k / 4) + \cos(\pi k / 3)$$

4.13. Try to find the spectrum of the following discrete non-periodic sequences.

(1) $x_1[k] = \alpha^k u[k] \quad |\alpha| < 1$ (2) $x_2[k] = \alpha^k u[-k] \quad |\alpha| > 1$

(3) $x_3[k] = \begin{cases} \alpha^{|k|} & |k| \leq M \\ 0 & \text{其他} \end{cases}$ (4) $x_4[k] = \alpha^k u[k+3] \quad |\alpha| < 1$

(5) $x_5[k] = \sum_{n=0}^{\infty} (1/4)^n \delta[k-3n]$ (6) $x_6[k] = \left(\frac{\sin(\pi k/3)}{\pi k} \right) \left(\frac{\sin(\pi k/4)}{\pi k} \right)$

4.14. We know the finite length sequence $x[k] = \{2, 1, -1, 0, 3, 2, 0, -3, -4\}$, try to determine the following expression value without calculating the DTFT of $x[k]$.

(1) $X(e^{j0})$ (2) $X(e^{j\pi})$ (3) $\int_{-\pi}^{\pi} X(e^{j\Omega}) d\Omega$

(4) $\int_{-\pi}^{\pi} |X(e^{j\Omega})|^2 d\Omega$ (5) $\int_{-\pi}^{\pi} \left| \frac{dX(e^{j\Omega})}{d\Omega} \right|^2 d\Omega$

4.15. We know the highest angular frequency of the signal is ω_m rad/s. If the following signal is sampled in time domain, try to determine the maximum sampling interval of its aliasing.

(1) $x(t/4)$ (2) $x^2(t)$ (3) $x(t) * x(t)$ (4) $x(t) + x(t/4)$

4.16. The spectrum of a complex signal is shown in figure 4-6. Try to plot the spectrum of signal with sampling angle frequency $\omega_s = \omega_m$.

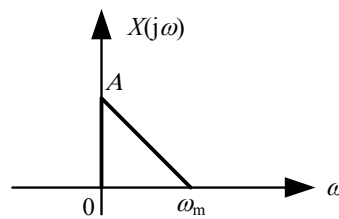


Figure.4-6

4.17. In the system shown in figure 4-7, the signal $x(t)$ is first bounded by an ideal low pass filter and then converted to a discrete sequence $y[k]$ through the A/D converter. Try to write out the spectrum expression and plot its spectrum.

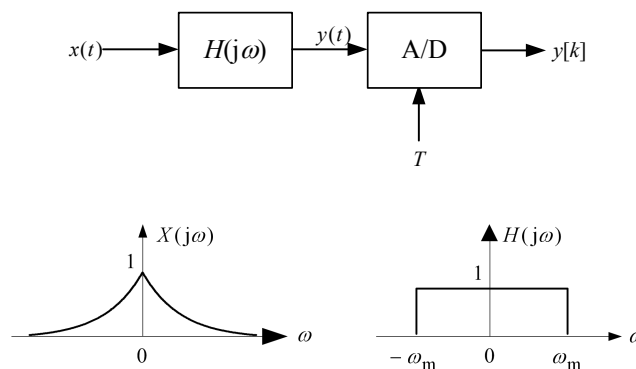


Figure.4-7

4.18. The spectrum of the finite length discrete sequence $x[k]$ is $X(e^{j\Omega})$, the discrete frequency spectrum $X[m]$ is obtained by sampling N points in one cycle, that

is $X[m] = X(e^{j\Omega}) \Big|_{\Omega = \frac{2\pi}{N}m}$. If the value of $x[k]$ is as follows, try to determine the time series

corresponding to the discrete spectrum, where $N=5$.

(1) $x[k] = \{1, 2, 3, 4\}$

(2) $x[k] = \{1, 2, 3, 4, 5, 6, 7\}$

4.19. Trigonometric wave signal $x(t)$ shown in figure 4-8.

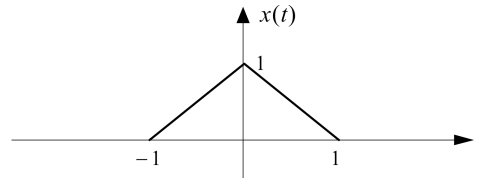


Figure.4-8

- (1) Try to calculate spectrum $X(j\omega)$ of signal $x(t)$, and draw the frequency spectrum.
- (2) The signal $x(t)$ is sampled equally at intervals of $T=0.1$ s, and the discrete sequence $x[k]$ is obtained. Write out the frequency spectrum $X(e^{j\Omega})$ of $x[k]$ and draw the spectrum diagram .
- (3) The signal $x(t)$ is cycled in $T_0=3$ s cycle, and the periodic signal $\tilde{x}_{T_0}(t) = \sum_{n=-\infty}^{\infty} x(t + 3n)$ is obtained. Try to obtain the frequency spectrum C_n of periodic signal $\tilde{x}_{T_0}(t)$ and draw the spectrum.
- (4) The periodic signals $\tilde{x}_{T_0}(t)$ are sampled equally at intervals of $T=0.1$ s, and the discrete periodic sequences $\tilde{x}[k]$ are obtained. Try to find the spectrum $\tilde{X}[m]$ of periodic sequences $\tilde{x}[k]$ and draw the frequency spectrum.